

An audit of the immrep25 unseen-peptide benchmark

Signal-to-noise of CDR3 homology and V/J chain pairing across the positive set

Anna E. Koneva

Mikhail Shugay*

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Abstract

We ask whether the *positive* TCR-peptide assignments in the immrep25 unseen-peptide benchmark carry epitope-specific signal, or only noise and publicity. Two biologically-grounded probes—within- vs. between-epitope CDR3 *homology* and per-epitope V/J *chain pairing*—are evaluated one-vs-many per epitope and aggregated across epitopes, then calibrated against cohorts of known quality (TCRvdb experimentally verified / refuted; VDJdb high- / low-reference-support) and two OLGA controls (raw random and generation-probability-matched) that fix the noise floor at $S/N=1$. On both probes and both chains the immrep25 positives sit just above that floor and one to three orders of magnitude below every quality cohort; their weak homology is fully explained by public TCRs (it collapses to the floor once public TCRs are removed) and their inter-chain pairing shows essentially no coupling. We conclude the positive labels carry no meaningful epitope-specific signal beyond noise and publicity.

1 Data and design

Cohorts. Every cohort is reduced to a common per-chain schema (one CDR3 with its V/J and epitope) and a paired schema ($\text{cdr3}\alpha, V\alpha, J\alpha, \text{cdr3}\beta, V\beta, J\beta, \text{epitope}$). The six cohorts span the expected signal-to-noise (S/N) range:

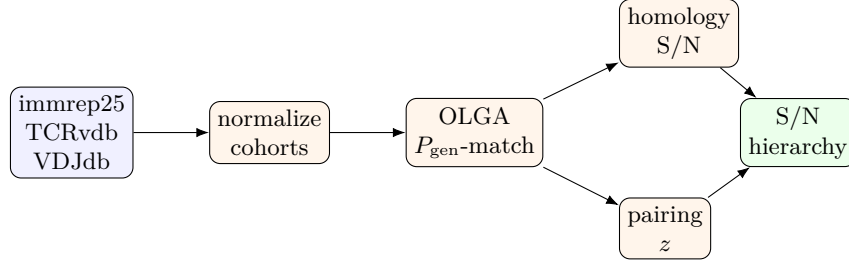
cohort	definition	role
TCRvdb true	TCRvdb, $p_{\text{adj}} < 10^{-5}$	experimentally verified (top)
VDJdb HQ	VDJdb human MHC-I, ≥ 2 references per (TCR, epitope)	strongly supported
VDJdb LQ	VDJdb human MHC-I, single reference	weakly supported
TCRvdb false	TCRvdb, $p_{\text{adj}} \geq 10^{-5}$	experimentally refuted
immrep25 pos	immrep25 <code>label=1</code>	system under test
OLGA pgen-matched	P_{gen} -matched to immrep25 pos	noise floor
OLGA random	raw OLGA generation	noise floor

Cohorts draw on VDJdb [1], the functionally-validated TCRvdb/MATCHMAKERS database [2], the immrep25 unseen-peptide benchmark [3], and an OLGA control built from the TCR generation model of Murugan et al. [4]. The immrep25 test set covers 20 nine-mer peptides that are *absent from VDJdb*, with 50 positives (and 450 negatives) each; only positives are audited. VDJdb HQ is defined per chain (TRA and TRB separately) as a (TCR, epitope) supported by ≥ 2 distinct references, with a paired complex counted HQ if *either* chain qualifies. TCRvdb true/false are dominated by the same two epitopes (YLQPRTFLL, GLCTLVAML), giving an epitope-matched verified-vs-refuted contrast.

*ISALGO lab. Correspondence: mikhail.shugay@gmail.com

One-vs-many. Every metric is computed *per epitope* against all others (each epitope with ≥ 30 records vs. the rest of its cohort) and then aggregated to a dataset value—geometric mean $\pm 95\%$ CI for homology S/N, mean $\pm 95\%$ CI for pairing—across epitopes. The signal therefore attaches to individual epitopes and the reported error bars are the epitope-to-epitope spread (n = number of qualifying epitopes); no bootstrap resampling is used. The two OLGA controls fix the noise floor: raw random draws and draws whose per-chain generation probability is matched to immrep25 (so chance matchability is controlled).

Pipeline.



2 Homology test

Metric. For a single chain we score each epitope one-vs-many. Only equal-length CDR3s can share a Hamming distance, so pairs are counted within length bins. For epitope e (N_e TCRs) against the other $N - N_e$ TCRs of its cohort,

$$P_{\text{self}} = \binom{N_e}{2}, \quad P_{\text{non}} = N_e (N - N_e)$$

(equal-length pairs); with $m_{\leq d}/l_{\leq d}$ the within-/against-rest pairs at Hamming $\leq d$ and an exposure-proportional pseudocount a (so $S/N \rightarrow 1$ under no enrichment),

$$S/N_e(d) = \frac{(m_{\leq d} + a)/P_{\text{self}}}{(l_{\leq d} + a P_{\text{non}}/P_{\text{self}})/P_{\text{non}}}.$$

The dataset value is the geometric mean of S/N_e over epitopes ($\pm 95\%$ CI); the error bars are thus the epitope-to-epitope spread. $d=0$ isolates publicity (identical CDR3s); $d=1$ is the substitution-neighbour convergence signal exploited by curated specificity databases [1]; signal $\Rightarrow S/N \gg 1$, noise $\Rightarrow S/N \approx 1$.

Result. Figure 1 orders the cohorts: VDJdb-HQ $>$ TCRvdb-true $>$ TCRvdb-false $>$ VDJdb-LQ $\gg$ immrep25 $>$ OLGA. On $TR\beta$ the verified cohorts reach $S/N = 1756$ (VDJdb-HQ) and 496 (TCRvdb-true), whereas immrep25 positives give only 2.69 and both OLGA controls sit at the floor (1.00 matched, 0.995 raw). The same holds on $TR\alpha$ (immrep25 3.88 vs. OLGA 0.932). immrep25’s CI excludes 1 but lies one to three orders of magnitude below every quality cohort. As the substitution radius grows (Fig. 2) the verified cohorts stay high while immrep25 and OLGA remain near the floor across $d = 1, 2, 3$.

By-epitope breakdown. Figure 3 plots the within-epitope neighbour rate for every epitope with $n \geq 30$ (one point per epitope; boxplots show the per-cohort quartiles); the vertical gap to each cohort’s non-self rate (red bar) is the S/N. immrep25’s points hug its non-self line, whereas the verified cohorts sit orders of magnitude above. Table 1 lists all 20 immrep25 epitopes: in total they contribute only 54 ($TR\beta$) / 53 ($TR\alpha$) within-epitope Hamming-1 neighbours — versus 0 / 1 for the P_{gen} -matched OLGA control and tens of thousands for the VDJdb cohorts — and even that handful is concentrated in a few epitopes (e.g. YLFNADIWI) and is entirely public (§4).

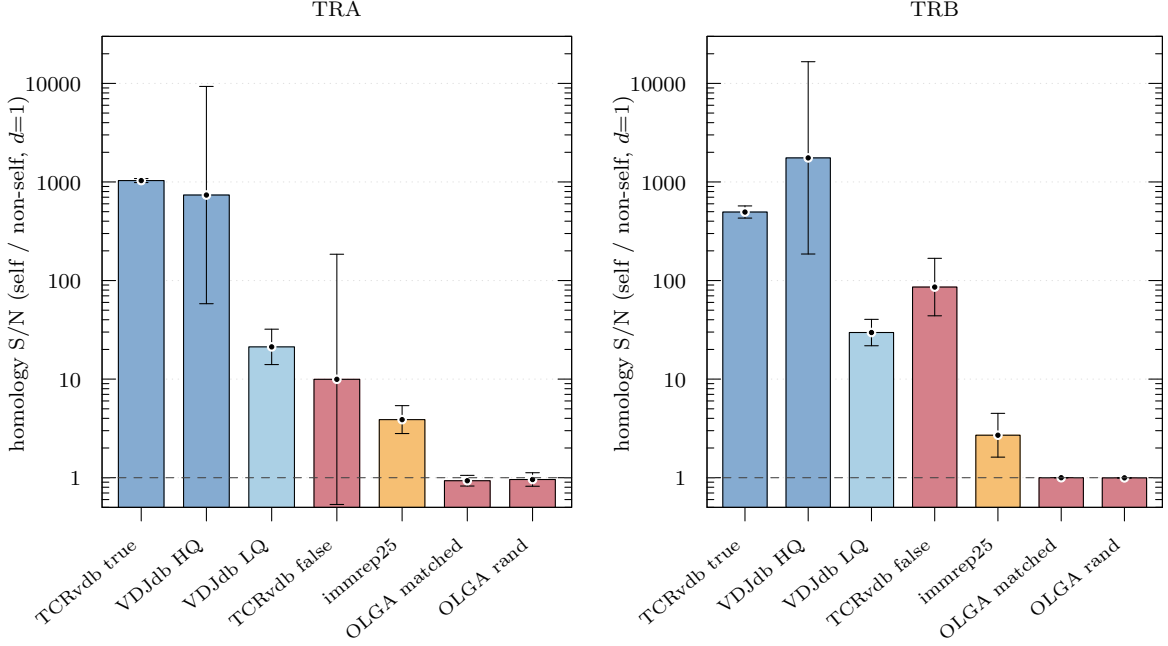


Figure 1: Homology S/N ($d=1$) per cohort, $TR\alpha$ and $TR\beta$ (one-vs-many: each epitope with ≥ 30 records vs. the rest; bar = geometric mean over epitopes, whiskers = 95% CI = between-epitope spread). Dashed line: noise floor S/N=1.

Graph view. The S/N ratio is the within- vs. between-epitope edge-density of the homology graph (nodes=CDR3, edges=Hamming ≤ 1 ; all epitopes with ≥ 30 records). Figure 4 renders it (graphviz `sfdp` layout): the verified cohorts form dense epitope-coloured cliques, while immrep25 and both OLGA controls are near-edgeless point clouds.

3 Pairing test

Metric. Genome-wide α/β pairing is close to random [5], but given an epitope the V/J usage may be constrained. For every epitope with ≥ 30 paired records we measure two signals as the *excess over a permutation null*, in bits, then average over epitopes ($\pm 95\%$ CI):

- **gene-usage bias:** $\overline{KL}(P_e \parallel Q)$ over axes $V\alpha, J\alpha, V\beta, J\beta$ (P_e = within-epitope gene distribution, Q = cohort background) minus its mean under size-matched random draws from Q .
- **inter-chain MI:** $I((V\alpha, J\alpha); (V\beta, J\beta))$ within epitope (Miller–Madow) minus its mean under $\alpha \leftrightarrow \beta$ shuffles (which keep the marginals but destroy coupling).

Both are 0 under no signal and grow (in bits) with genuine per-epitope structure.

Result. Figure 5 shows both. The verified cohorts carry substantial gene-usage bias (VDJdb-HQ 1.62 bits) and real inter-chain coupling (TCRvdb-true 1.04 bits). immrep25 positives are near the floor on both: gene-usage bias 0.203 bits and, decisively, an inter-chain coupling of only 0.047 bits—an order of magnitude below every quality cohort and barely above the OLGA controls (0.006 / -0.000 bits). The residual gene-usage bias is consistent with the public, HLA-restricted composition of the positives rather than epitope-specific pairing.

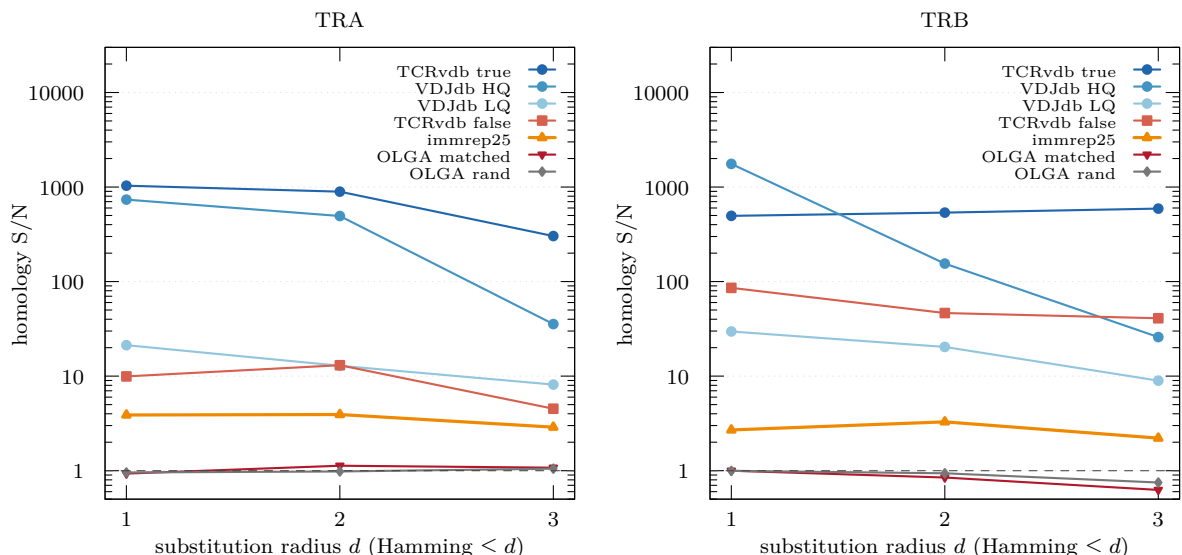


Figure 2: Homology S/N vs. substitution radius $d = 1, 2, 3$ (one-vs-many geometric mean). Verified cohorts stay high; immrep25 and both OLGA controls track the floor.

4 Publicity control

The one alternative to “pure noise” is *publicity*: immrep25 might reuse public TCRs whose chance homology mimics epitope signal. Although only 0.3% of positives share *both* chains exactly with VDJdb, allowing a single substitution and either chain, 74.5% of immrep25 positives are within $\text{Hamming} \leq 1$ of a known VDJdb/TCRvdb TCR (61.5% on α , 31.2% on β)—the α chain being especially germline-public. Consistently, the positives are enriched for high generation probability relative to raw OLGA sampling (Fig. 6), i.e. they are “easy-to-generate”, public-leaning sequences; this is exactly why the random control must be P_{gen} -matched.

Removing public positives collapses the (already weak) homology to the noise floor: one-vs-many S/N falls from 2.69 to 1.00 on $\text{TR}\beta$ and from 3.88 to 1.37 on $\text{TR}\alpha$ (Fig. 7). The residual signal was therefore publicity, not epitope-specific convergence.

5 Conclusion

Across both probes and both chains the calibration cohorts recover the expected order, and the immrep25 positives fall at the bottom, just above the two OLGA random floors and one to three orders of magnitude below every quality cohort (Fig. 8). Their weak homology is contributed by public TCRs and collapses to the floor when those are removed; their inter-chain pairing shows essentially no coupling. We therefore find *no meaningful epitope-specific signal—in homology or chain pairing—among the immrep25 positives beyond noise and publicity*, consistent with the benchmark’s own finding that models do not exceed random guessing on unseen peptides [3].

Caveats. TCRvdb true/false separate less than expected because p_{adj} scores differential enrichment in one screen rather than absence of specificity, so “false” still contains epitope-associated TCRs; and TCRvdb and VDJdb-HQ rest on only 2 and 5 epitopes respectively, so their absolute S/N is uncertain (wide CIs). Neither affects immrep25, whose 20 epitopes place it just above the OLGA floor on every probe. Reproduce with `bash scripts/fetch_vdjdb.sh; python src/build_olga.py; python run_audit.py; make`.

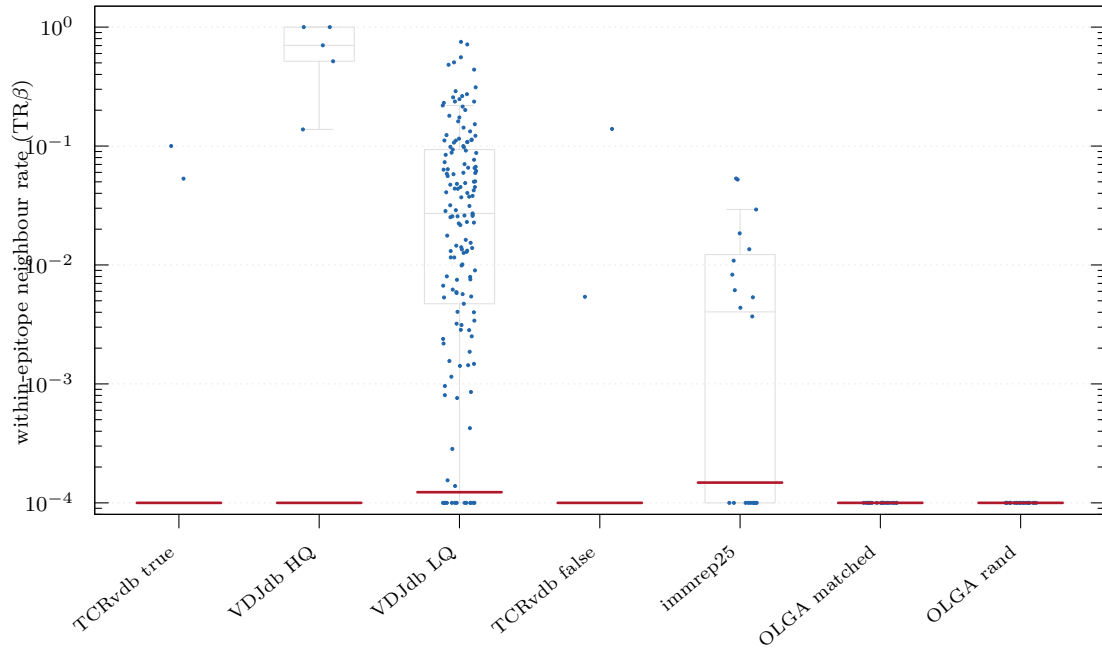


Figure 3: Per-epitope within-epitope neighbour rate ($TR\beta$, $\text{Hamming} \leq 1$; each point is one epitope, $n \geq 30$; boxplots = per-cohort quartiles). Red bars = cohort pooled cross-epitope (non-self) rate; the vertical gap is the homology S/N.

References

- [1] Mikhail Shugay, Dmitriy V Bagaev, Ivan V Zvyagin, Renske M Vroomans, Jeremy Chase Crawford, Garry Dolton, Ekaterina A Komech, Anastasiya L Sycheva, Anna E Koneva, Evgeniy S Egorov, Alexey V Eliseev, Ewald Van Dyk, Pradyot Dash, Meriem Attaf, Cristina Rius, Kristin Ladell, James E McLaren, Katherine K Matthews, E Bridie Clemens, Daniel C Douek, Fabio Luciani, Debbie van Baarle, Katherine Kedzierska, Can Kesmir, Paul G Thomas, David A Price, Andrew K Sewell, and Dmitriy M Chudakov. Vdjdb: a curated database of t-cell receptor sequences with known antigen specificity. *Nucleic Acids Research*, 46(D1):D419–D427, September 2017.
- [2] Marius Messemaker, Bjørn P.Y. Kwee, Živa Moravec, Daniel Álvarez-Salmoral, Jos Urbanus, Sam de Paauw, Jeroen Geerligs, Rhianne Voogd, Ben Morris, Aurélie Guislain, Maike Mußmann, Yaël Winkler, Maxime Steinmetz, Matyas Iras, Eric Marcus, Jonas Teuwen, Anastassis Perrakis, Roderick L. Beijersbergen, Wouter Scheper, and Ton N. Schumacher. A functionally validated tcr-pmhc database for tcr specificity model development. *bioRxiv*, May 2025.
- [3] Eve Richardson, Yannick Jurriaan Maria Aarts, John A. Altin, Coos A. B. Baakman, Philip Bradley, Binbin Chen, Joakim Clifford, Manjima Dhar, Danielle Diepenbroek, Ethan Fast, Ragul Gowthaman, Jieling He, Vadim Karnaukhov, Dario F. Marzella, Pieter Meysman, Morten Nielsen, Jonas Birkelund Nilsson, Sebastian Nymann Deleuran, Farzaneh M. Parizi, Aurelien Pelissier, Brian G. Pierce, Maria Rodriguez Martinez, Dona Roran A R, Shayana Saravanakumar, Yanjun Shao, Nils Smit, Max Van Houcke, Gian Marco Visani, Yat-Tsai Richie Wan, Xiaowo Wang, Lawson Woods, Sander Wuyts, Chengkai Xiao, Li C. Xue, Justin Barton, Matthew Noakes, Damon H. May, Bjoern Peters, and IMMREP25 Participant Consortium. Immrep25: Unseen peptides. *bioRxiv*, 2026.
- [4] Anand Murugan, Thierry Mora, Aleksandra M. Walczak, and Curtis G. Callan. Statistical inference of the generation probability of t-cell receptors from sequence repertoires. *Proceedings*

epitope	n	β nbrs	α nbrs
YLFNADIWI	50	20	1
MTDYDYLEV	50	13	3
ILHTHVPEV	50	5	7
TVYPYGTSL	50	5	2
YMFYDGYDV	50	3	2
REDDYSVWL	50	2	1
SQFNWTIYL	50	2	1
GEVLACYAL	50	1	1
FLDCKSYIL	50	1	4
FEAQPGALL	50	1	3
SEESAFYVL	50	1	2
ILMHATYFL	50	0	1
YEDGVIFYL	50	0	5
MENWSALEL	50	0	0
HLDDYPYLM	50	0	1
KLYPFLWFA	50	0	11
YENGSTPVL	50	0	1
GENALTYAL	50	0	3
MEMPDYLLL	50	0	3
YESYIPGAL	50	0	1
immrep25 total	1000	54	53
OLGA pgen-matched	999	0	1

Table 1: immrep25 positives by epitope (n TCRs) and their count of within-epitope Hamming-1 CDR3 neighbours per chain, versus the OLGA control. Sorted by $\text{TR}\beta$.

of the National Academy of Sciences, 109(40):16161–16166, September 2012.

- [5] Dmitrii S. Shcherbinin, Vlad A. Belousov, and Mikhail Shugay. Comprehensive analysis of structural and sequencing data reveals almost unconstrained chain pairing in $\text{tcr}\alpha\beta$ complex. *PLOS Computational Biology*, 16(3):e1007714, March 2020.

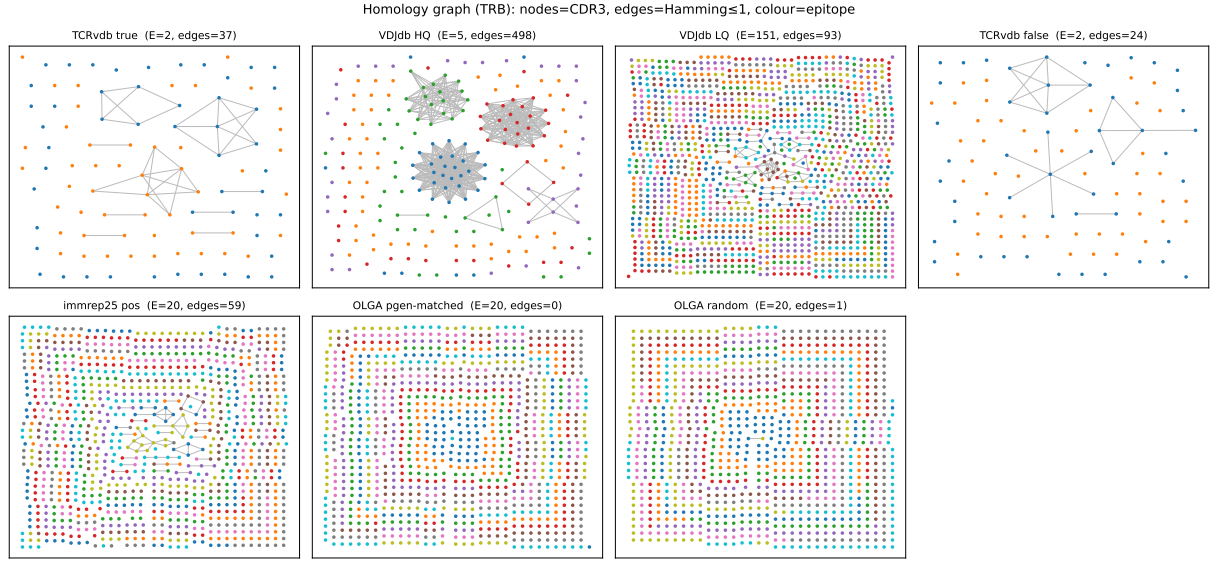


Figure 4: TR β homology graph per cohort (all epitopes ≥ 30 records). Verified cohorts form dense epitope-coloured cliques; immrep25 and OLGA are near-edgeless.

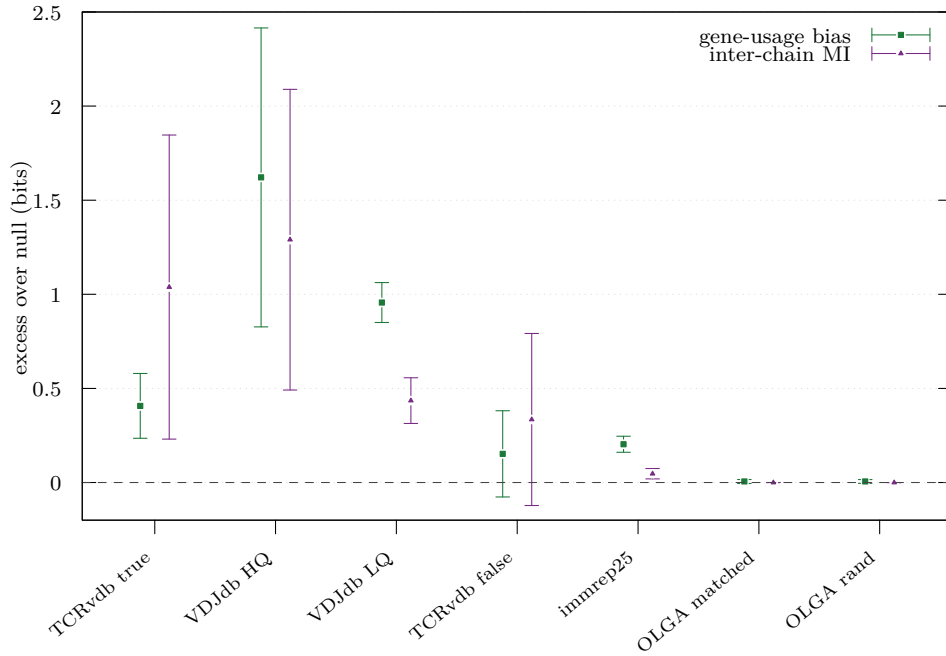


Figure 5: Pairing signals per cohort (one-vs-many over all epitopes with ≥ 30 records): per-epitope excess over the permutation null, in bits (mean $\pm$ 95% CI). Squares = gene-usage bias, triangles = inter-chain MI. Dashed line = 0 (random). immrep25 and both OLGA controls sit at the random line, especially for MI.

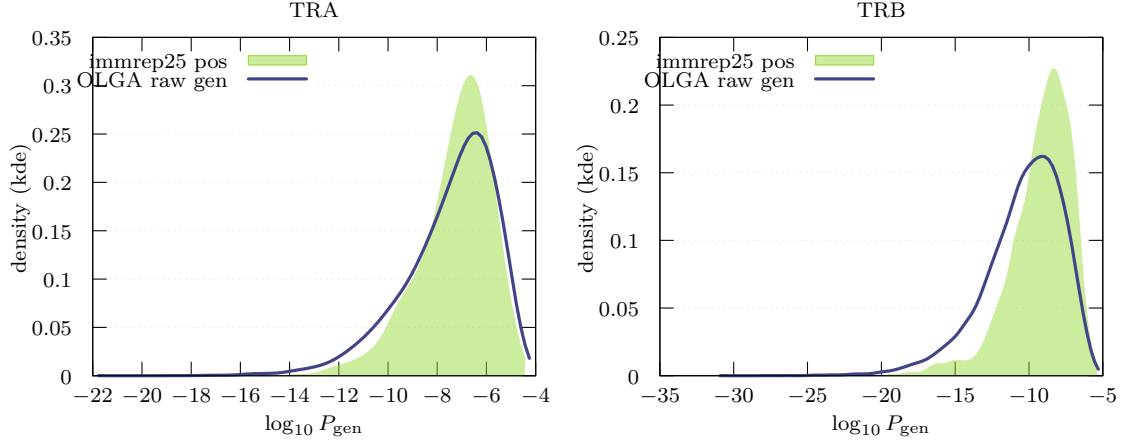


Figure 6: Per-chain $\log_{10} P_{\text{gen}}$ (kernel density): immrep25 positives (filled) vs. raw OLGA generation (line). Positives skew to higher P_{gen} (public-leaning), motivating the P_{gen} -matched control.

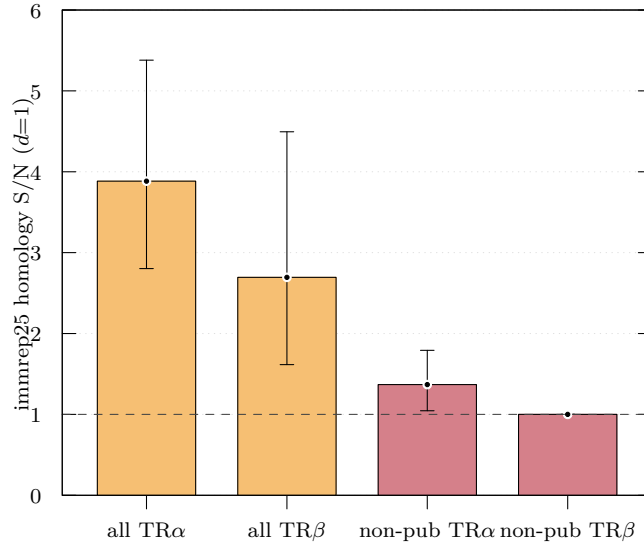


Figure 7: immrep25 homology S/N ($d=1$) for all positives vs. non-public positives (Hamming ≤ 1 to references removed). Removing public TCRs drives S/N to the noise floor (dashed line).

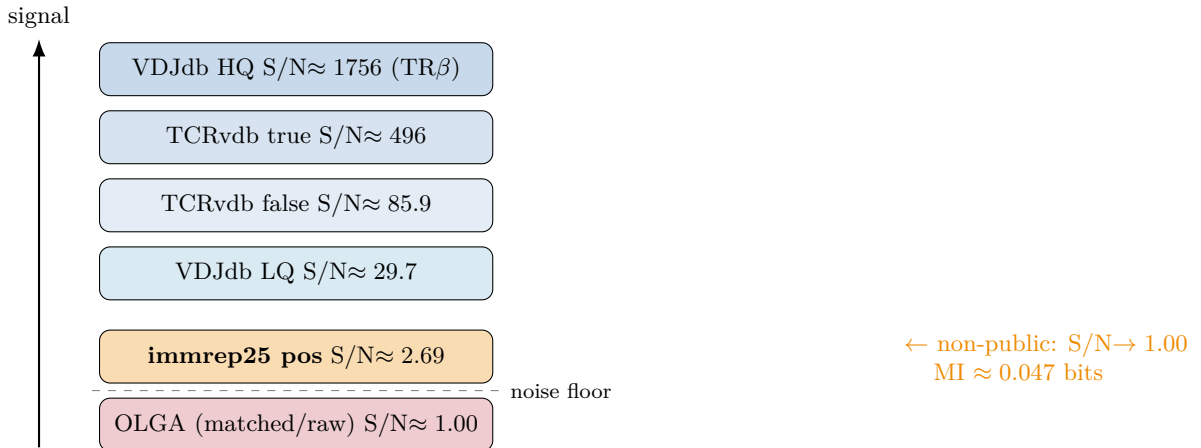


Figure 8: Signal-to-noise ladder (TR β homology, one-vs-many). The immrep25 positives sit one rung above the random floor; stripping public TCRs lowers them onto it.